

Challenge (Habanero)

- Q1.** What is the factored form of $x^3 + 8$?
(A) $(x + 2)(x^2 - 2x + 4)$
(B) $(x + 2)(x^2 + 2x + 4)$
(C) $(x + 2)(x^2 - 2x - 4)$
(D) $(x - 2)(x^2 + 2x + 4)$
(E) $(x - 2)(x^2 - 2x - 4)$
- Q2.** A hiker is traveling a distance of $x^3 - 27$ miles. What is the factored form of the distance?
(A) $(x - 3)(x^2 + 3x + 9)$
(B) $(x + 3)(x^2 - 3x + 9)$
(C) $(x - 3)(x^2 - 3x - 9)$
(D) $(x + 3)(x^2 + 3x - 9)$
(E) $(x + 3)(x^2 - 3x + 9)$
- Q3.** What is the factored form of $x^3 - 64$?
(A) $(x - 4)(x^2 + 4x + 16)$
(B) $(x + 4)(x^2 - 4x + 16)$
(C) $(x - 4)(x^2 - 4x - 16)$
(D) $(x + 4)(x^2 + 4x + 16)$
(E) $(x + 4)(x^2 - 4x - 16)$
- Q4.** A wildlife reserve has a population of $x^3 + 125$ animals. What is the factored form of the population?
(A) $(x + 5)(x^2 - 5x + 25)$
(B) $(x - 5)(x^2 + 5x + 25)$
(C) $(x + 5)(x^2 + 5x - 25)$
(D) $(x - 5)(x^2 - 5x + 25)$
(E) $(x - 5)(x^2 + 5x - 25)$
- Q5.** What is the factored form of $x^3 - 1$?
(A) $(x - 1)(x^2 + x + 1)$
(B) $(x + 1)(x^2 - x + 1)$
(C) $(x - 1)(x^2 - x - 1)$
(D) $(x + 1)(x^2 + x - 1)$
(E) $(x + 1)(x^2 + x + 1)$
- Q6.** A fuel tank can hold $x^3 + 216$ gallons of fuel. What is the factored form of the capacity?
(A) $(x + 6)(x^2 - 6x + 36)$
(B) $(x - 6)(x^2 + 6x + 36)$
(C) $(x + 6)(x^2 + 6x - 36)$
(D) $(x - 6)(x^2 - 6x + 36)$
(E) $(x - 6)(x^2 + 6x - 36)$
- Q7.** What is the factored form of $x^3 + 343$?
(A) $(x + 7)(x^2 - 7x + 49)$
(B) $(x - 7)(x^2 + 7x + 49)$
(C) $(x + 7)(x^2 + 7x - 49)$
(D) $(x - 7)(x^2 - 7x + 49)$
(E) $(x - 7)(x^2 + 7x - 49)$
- Q8.** A park has a trail of length $x^3 - 1000$ feet. What is the factored form of the length?
(A) $(x - 10)(x^2 + 10x + 100)$
(B) $(x + 10)(x^2 - 10x + 100)$
(C) $(x + 10)(x^2 + 10x - 100)$
(D) $(x - 10)(x^2 - 10x + 100)$
(E) $(x - 10)(x^2 + 10x - 100)$

Open Ended Questions (FRQ)

- Q9.** A conservation group is monitoring the population of $x^3 - 512$ animals in a wildlife reserve. Factor the expression and find the value of x if the population is 128 animals.
- Q10.** A fuel-efficient vehicle can travel a distance of $x^3 + 1000$ miles on a single tank of gas. Factor the expression and find the value of x if the vehicle travels 10 miles.

Chef's Tips

- Provide feedback on the factoring process
- Emphasize the importance of checking the factored form
- Encourage students to use real-world applications to justify their answers